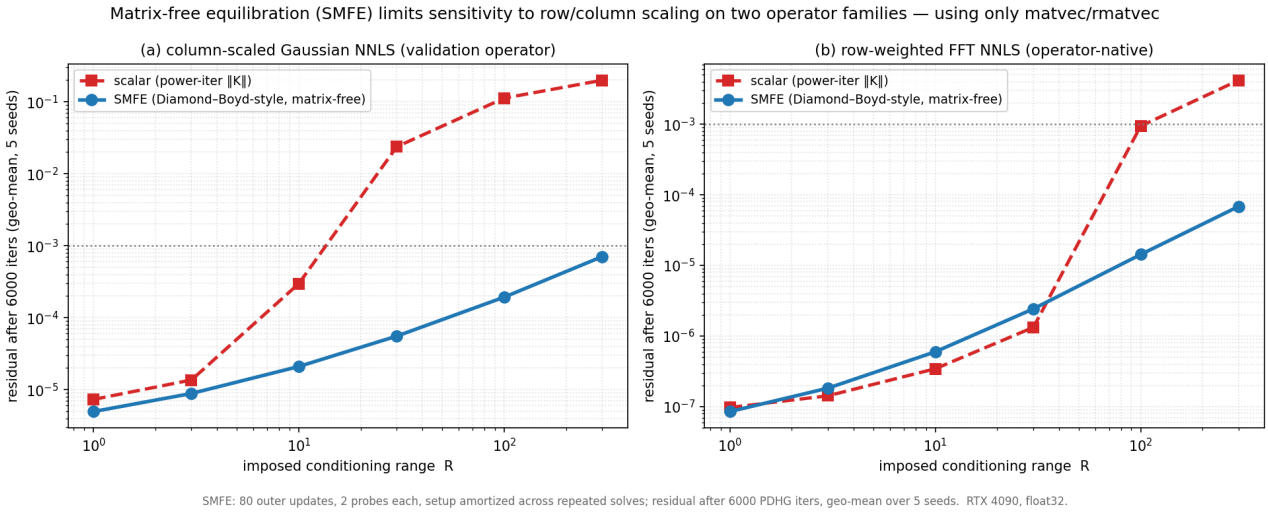
1. Matrix-free equilibration

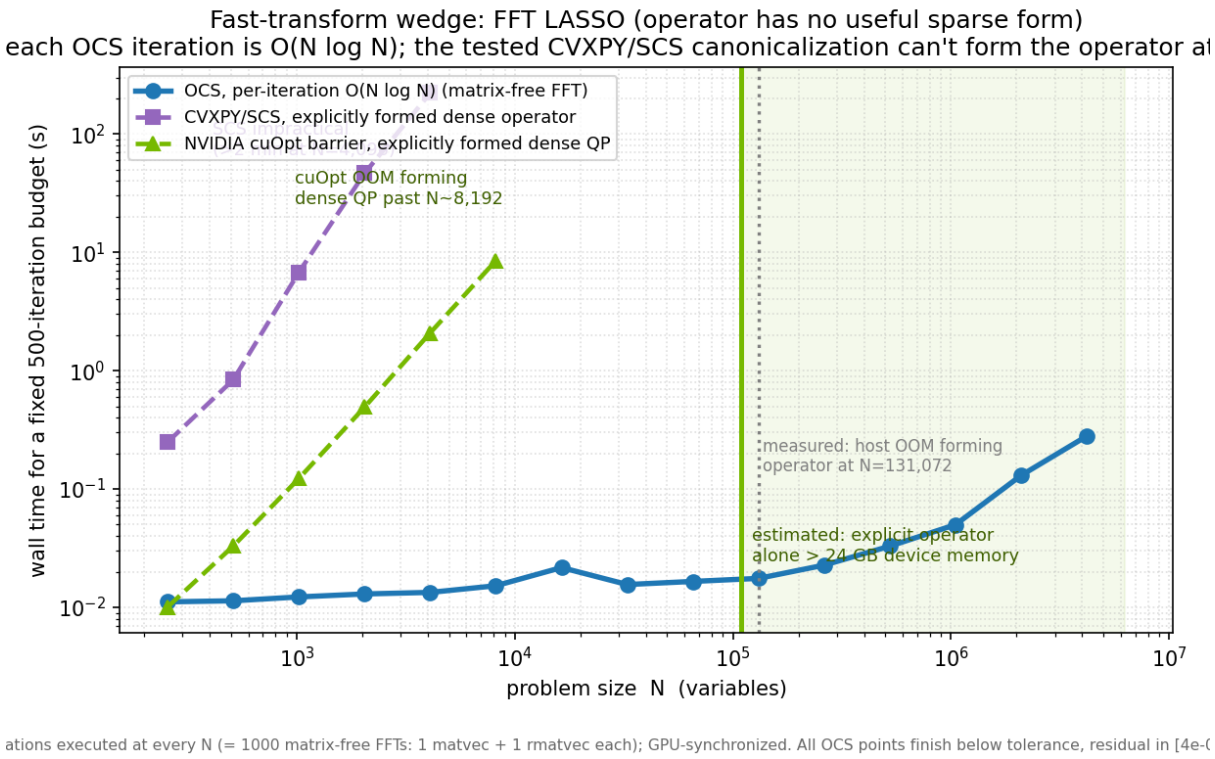


Problem. Two nonnegative least-squares families: $\min \ \frac{1}{2}\|Ax-b\|_2^2$ subject to $x \geq 0$.

How the data was generated. Group (a) column-scaled Gaussian: a base matrix $G \in \mathbf{R}^{240\times 180}$ is drawn once with i.i.d. $\mathcal{N}(0,1)$ entries then each instance multiplies its columns by $s_j = R^{u_j}$, $u_j \sim U(0,1)$ (so column norms span $[1,R]$), giving $A = G\,\mathrm{diag}(s)$; the right-hand side is $b = Ax_{\mathrm{true}}$ with $x_{\mathrm{true}} = |\mathcal{N}(0,1)|$ entrywise. Group (b) row-weighted FFT: $K = \mathrm{diag}(w)F$, where F is a fixed 50%-subsampled length-1024 DFT and the row weights are $w_i = R^{u_i}$, $u_i \sim U(0,1)$; x_{true} is 5%-sparse with $|\mathcal{N}(0,1)|$ on its support, and $b = Kx_{\mathrm{true}}$. Five seeds per point, and the conditioning parameter R is swept over $\{1,3,10,30,100,300\}$.

Plot. x-axis: imposed conditioning range R; y-axis: residual reached after a fixed 6000-iteration budget (geometric mean over 5 seeds), for plain scalar step sizes vs. the stochastic matrix-free equilibration. Scalar steps blow up as R grows; equilibrated steps stay near tolerance

2. FFT wedge

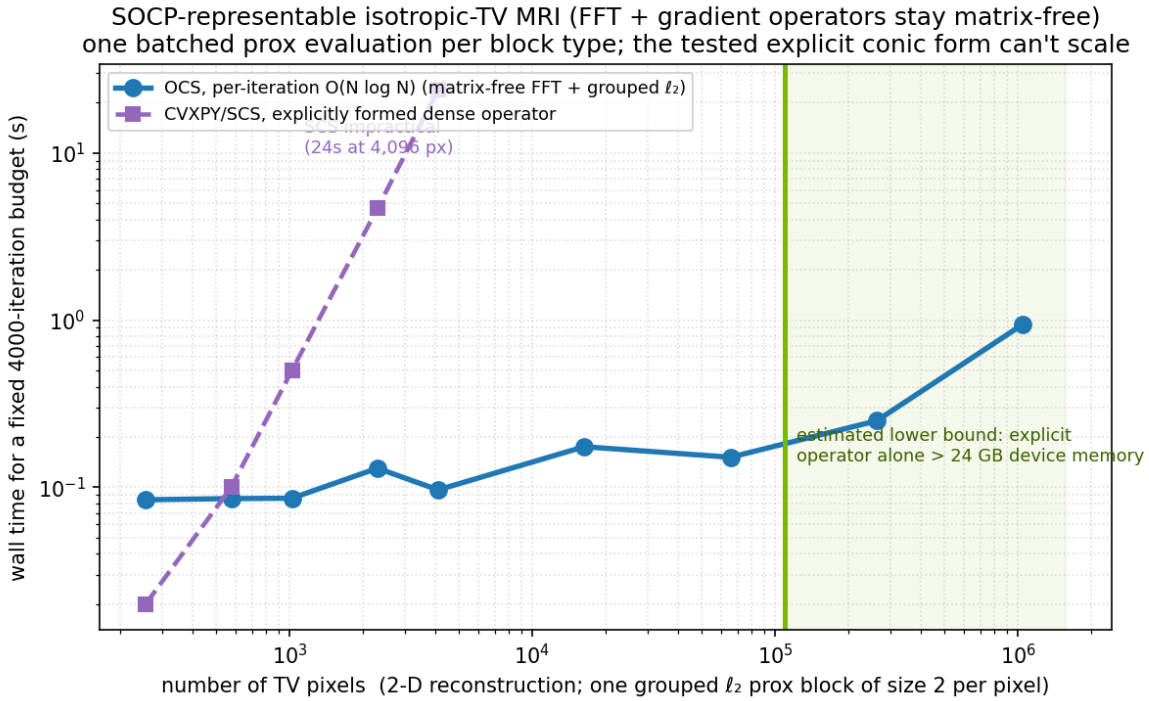


Problem. Compressed-sensing LASSO: $\min \frac{1}{2}\|MFx - y\|_2^2 + \lambda\|x\|_1$, $\lambda = 10^{-3}$. $K = MF$ is a subsampled DFT (M a 25% random row mask, F the FFT).

How the data was generated. K is built by `FFTMask.random(n, 0.25, seed=0)`: the length-n FFT with a fixed random 25% of the output rows kept. y is a single fixed i.i.d. $\mathcal{N}(0,1)$ vector. The plot measures runtime/memory scaling, not recovery accuracy. Signal length n is swept $256 \to 4 \times 10^6$.

Plot. x-axis n; y-axis wall time for a fixed 500-iteration budget (implementation scaling, not time to a common accuracy). The green marks where the formed dense operator exceeds the 24GB GPU; cuOpt's barrier on the equivalent QP runs out of memory forming $K^\top K$ near $n \approx 16{,}000$.

3. MRI / SOCP wedge



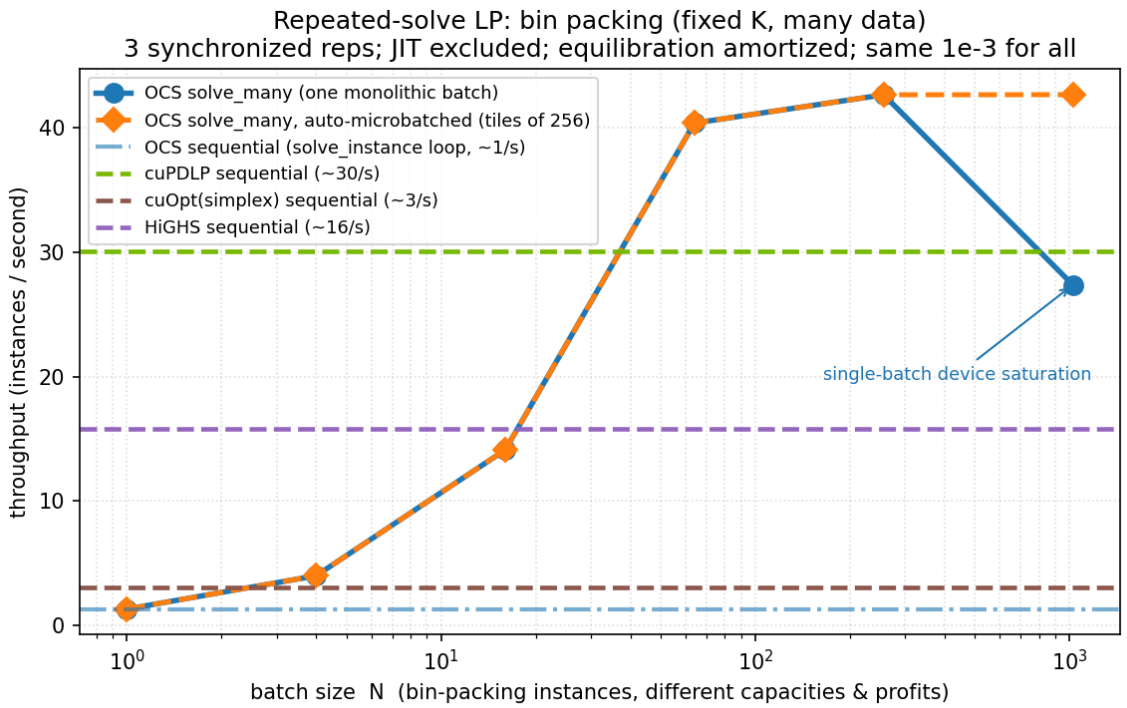
d budget: exactly 4000 iterations executed at every size, GPU-synchronized. All OCS points finish below tolerance, residual in [3e-07, 3e-04]. RTX 4090, floa

Problem. Isotropic TV image reconstruction: $\min \frac{1}{2}\|Fx - y\|_2^2 + \lambda \sum_i \|(Dx)_i\|_2$, $\lambda = 0.03$. F is a 35%-subsampled 2-D DF

How the data was generated. $F = \texttt{FFTMask.random}(L^2, 0.35, \text{seed} = 0)$; $D = \texttt{Gradient2D}(L, L)$. y is one fixed i.i.d. $\mathcal{N}(0, 1)$ vector (seed 0). Image side L is swept $16 \to 1024$ (up to 10^6 pixels).

Plot. SOCP: x-axis number of pixels, y-axis wall time for a fixed 4000-iteration budget; the green is where an explicit conic canonicalization would exceed device memory.

4. Batched throughput



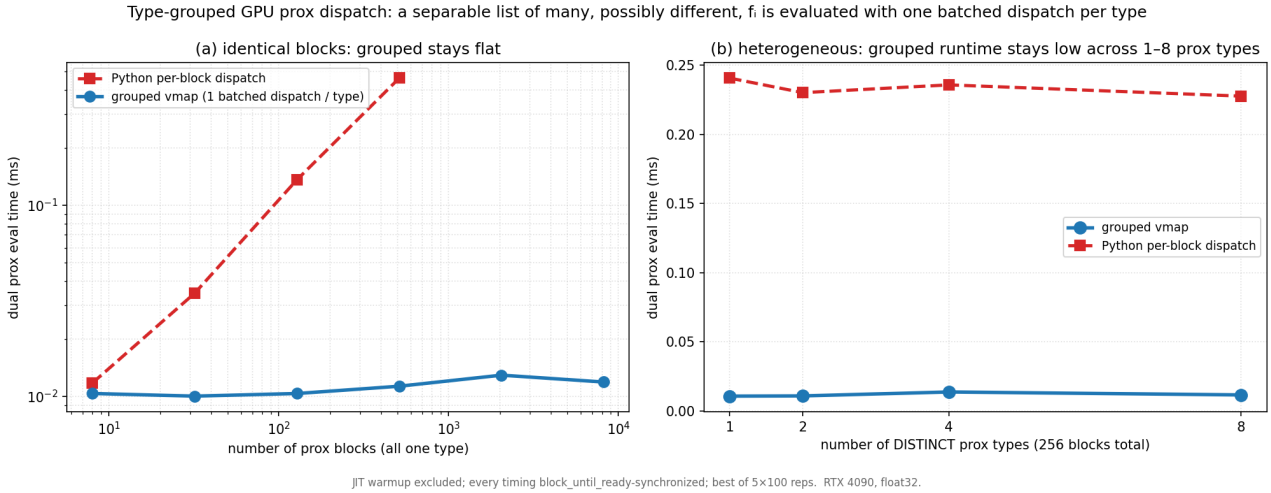
4,800 variables/instance (120 items × 40 bins). RTX 4090, float32.

Problem. Bin packing: $I = 120$ items into $B = 40$ bins under $R = 3$ resource constraints, 4,800 variables per instance.

How the data was generated. The resource-usage table $R_{\text{mat}} \in \mathbf{R}^{3\times 120}$ is drawn once with entries $U(0.1, 1)$ and shared by every instance (this is the fixed operator). Per instance (a separate stream, seed 4) the profits are $w_i \sim U(0.5, 1.5)$, and each bin's capacity is $C_{rb} = 0.6 \cdot \left(\sum_i R_{\text{mat}}[r, i]/B\right) \cdot U(0.7, 1.3)$. The batch is N such instances; only w and C vary.

Plot. x-axis number of instances N; y-axis throughput (instances/sec). Curves are: OCS as one batched `solve_many`, OCS auto-microbatched into tiles of 256, OCS solved one at a time, and cuOpt / HiGHS on the formed LP one at a time. Batching the whole family into a single GPU solve beats sequential.

5. Grouped prox dispatch

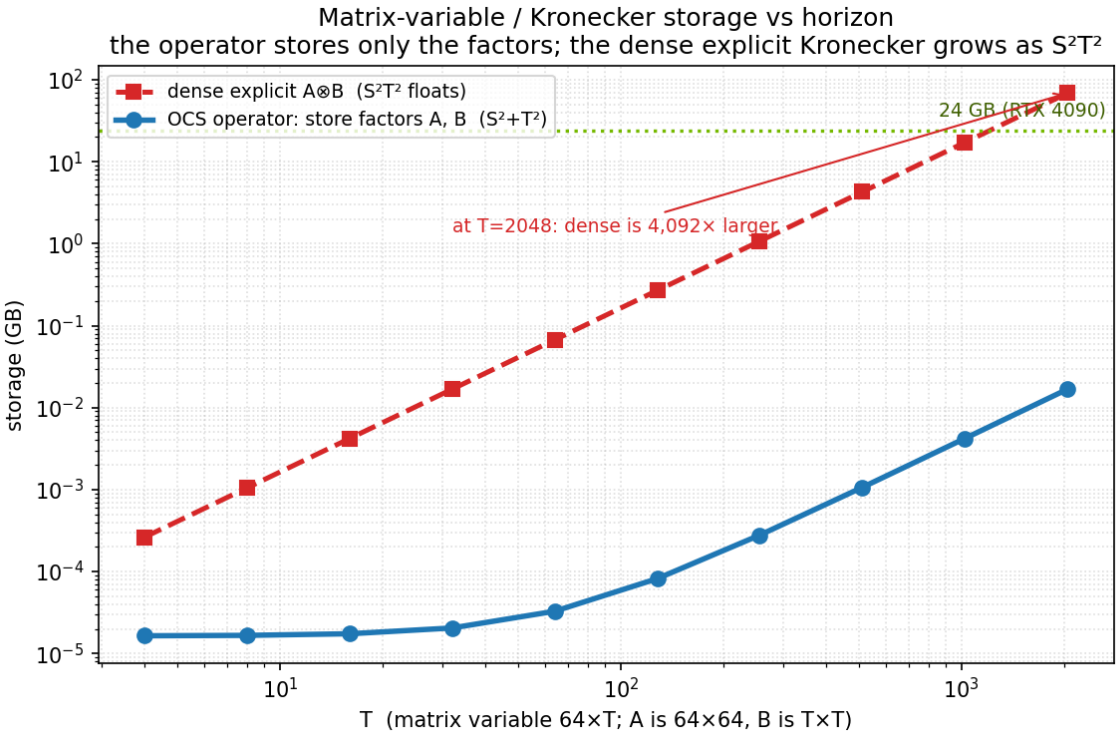


Plot. Left side (a): evaluate B identical size-8 second-order-cone prox blocks, grouped `vmap` dispatch vs. a Python per-block loop, as B grows (grouped stays flat; the loop is $\sim 46\times$ slower at 512 blocks). Right side (b): hold the total block count fixed and grow the number of distinct block types $1 \to 8$.

How the data was generated. The inputs are just arbitrary vectors fed to the prox. Plot (a) constructs B identical size-8 SOC blocks; Plot (b) fixes the total block count and varies the number of distinct block types (1, 2, 4, 8 drawn from SOC, box, nonneg, ℓ_1, ...). Times are best of 5×100 reps.

Conclusion. The prox stage is flat in the number of blocks and cheap across heterogeneous mixes, because each block type is one batched dispatch.

6. Kronecker storage (storage comparison, no solve)



Plot. The operator $K = A \otimes B$, applied to $\mathrm{vec}(X)$ as $\mathrm{vec}(AXB^\top)$, with A as $S \times S = 64 \times 64$ and B being $T \times T$. Plots the memory to store the dense explicit $A \otimes B$ ($(ST)^2$ floats) vs. storing only the two factors A, B ($S^2 + T^2$).

How the data was generated. The curves are exact byte counts as a function of T: the dense explicit $A \otimes B$ needs $(ST)^2$ floats, versus $S^2 + T^2$ to store just the factors. T is swept $4 \to 2048$. A pure memory comparison, not a measured run.

Conclusion. Never forming the operator is the difference between $O(S^2T^2)$ and $O(S^2 + T^2)$ memory: the dense form crosses 24 GB at $T \approx 1210$, and at $T = 2048$ it is $\sim 4{,}000\times$ the factors' footprint.

7. Reliability Sweep

OCS reliability battery (float32, RTX 4090)
›orted families: 45/45 agree with the reference (100%). 'reached tol' = residual < 0.001 (OCS termination); 'agrees w/ ref' = obj rel-err < 0.01 vs CVXPY / exact projec

problem class	trials	reached tol	agrees w/ ref	med rel-err	worst rel-err
SUPPORTED (end-to-end solve)					
LP min $c^T x$, $Ax=b$, $x\geq 0$	9	9/9 (100%)	9/9 (100%)	2.4e-04	9.1e-04
QP box least-squares	9	9/9 (100%)	9/9 (100%)	3.0e-07	5.1e-07
SOCP min $c^T x$, $\|Ax\|_2 \leq 1$	9	9/9 (100%)	9/9 (100%)	7.9e-04	9.3e-03
Quantile regression	9	9/9 (100%)	9/9 (100%)	1.7e-06	7.0e-06
Logistic regression	9	9/9 (100%)	9/9 (100%)	7.1e-08	1.6e-07
EXPERIMENTAL (cone-block projection)					
PSD-cone projection	9	9/9 (100%)	9/9 (100%)	3.0e-06	1.3e-05
Exp-cone projection	9	9/9 (100%)	9/9 (100%)	1.9e-04	4.4e-04
Power-cone projection	9	9/9 (100%)	9/9 (100%)	7.4e-05	9.7e-05

Problem. A sweep of classes: LP, QP, SOCP, quantile regression, logistic regression, and SDP/exp/power cone projections. Every class uses seed $= 100 \cdot (\text{size}) + \{0, 1, 2\}$: three seeds per size, three sizes per class; each instance is checked against CVXPY (SCS/Clarabel).

How the data was generated (per class).
1) LP: $\min c^\top x$ s.t. $Ax = b$, $x \geq 0$; $A \in \mathbf{R}^{m \times 2m}$ i.i.d. $\mathcal{N}(0,1)$, feasible point $x_{\text{true}} = |\mathcal{N}(0,1)|$, $b = Ax_{\text{true}}$, $c = |\mathcal{N}(0,1)|$; with $m = 8, 16, 32$.
2) QP: $\min \frac{1}{2}\|Ax - b\|_2^2$ s.t. $0 \leq x \leq 1$; $A \in \mathbf{R}^{2n \times n}$ i.i.d. $\mathcal{N}(0,1)$, $b \sim \mathcal{N}(0,1)$; with $n = 8, 16, 32$.
3) SOCP: $\min c^\top x + \frac{\mu}{2}\|x\|_2^2$ s.t. $\|Ax\|_2 \leq 1$, $\mu = 10^{-2}$; $c \sim \mathcal{N}(0,1)$, $A \in \mathbf{R}^{n \times n}$ i.i.d. $\mathcal{N}(0,1)$; with $n = 6, 12, 24$.
4) Quantile ($\tau = 0.75$): $\min \sum \text{pinball}_\tau(y - X\theta) + \frac{\mu}{2}\|\theta\|_2^2$, $\mu = 10^{-2}$; $X \in \mathbf{R}^{20n \times n}$ i.i.d. $\mathcal{N}(0,1)$, true $\theta \sim \mathcal{N}(0,1)$, $y = X\theta + 0.3\, t_3$; with $n = 4, 8, 16$.
5) Logistic: $\min \sum \log(1 + \exp(-b_i x_i^\top \theta)) + \frac{\mu}{2}\|\theta\|_2^2$, $\mu = 0.1$; $X \in \mathbf{R}^{20n \times n}$ i.i.d. $\mathcal{N}(0,1)$, true $\theta \sim \mathcal{N}(0,1)$, labels $b_i = \text{sign}(x_i^\top \theta + 0.5\,\mathcal{N}(0,1))$; with $n = 4, 8, 16$.
6) SDP/exp/power (projection validation): $\min \frac{1}{2}\|x - v\|_2^2$ s.t. x in the cone, v a random Gaussian point (symmetrized Gaussian matrix for PSD, $v \sim \mathcal{N}(0,1)$ for exp/power); checked against the exact projection (eigenvalue clip for PSD) or SCS.

Table. Per class: "reached tolerance" (OCS residual $< 10^{-3}$) and "agrees with reference" (objective relative error $< 10^{-2}$). 45 instances pass both.